

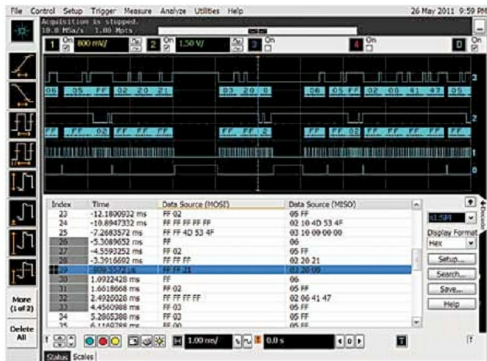


Fig. 3: MSOs can use any combination of digital or analogue channels for protocol decode as shown in this SPI decode example. This preserves some analogue channels for other time-correlated measurements, and allows sufficient channel availability if needs arise to decode multiple buses simultaneously

are working with. See what types of packet-based triggering is available and whether the trigger is implemented in hardware or software.

5 How much memory does your scope support for packet capture?

Users typically don't know how much memory is needed until they encounter a situation that requires longer time capture than their scope is capable of. Scopes typically come with some base memory, which can optionally be expanded by purchasing a licence to turn on additional memory. Scopes acquire asynchronously and hence use memory more quickly than dedicated protocol analysers or state-based logic analysers. For this reason, protocol apps benefit greatly from deep-memory scopes.

Check whether the scope under consideration allows you to independently set memory depth, sample rate and time base. This capability makes it dramatically easier to capture and view protocol signals with full memory depth. On scopes where this is not possible, users must capture the entire acquisition on screen, then pan and zoom to see packet detail.

Serial traffic often incorporates periods of dense activity followed by relatively long periods of dead time. For this purpose, using the scope's segmented memory mode enables users to capture significantly longer periods with the same amount of memory. Each segment is started when the scope sees a specified trigger condition. For example, when a USB device sends a number of packets, each with a setup packet, use of segmented memory allows the users to capture this sequence of events using a hundred times less memory (50 kpts instead of 5 Mpts).

6 How fast does your scope process and display serial packets?

Decoding serial packets can be taxing on scope performance. While vendors will

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tout fast analogue channel update rates when very shallow memory is used, few vendors will communicate how fast or slow they decode serial packets—particularly with deep memory.

For example, one scope vendor specifies a maximum update rate of 250 thousand waveforms per second. With 10 Mpts used for serial decode, the same scope updates at just one waveform every six seconds—more than a million times slower.

Slow update rates for protocol are problematic for several reasons. When viewing signals in real time, slow update rates dramatically degrade signal detail seen on the scope, miss showing infrequent events and frustrate users with sluggish instrument control.

7 What benefits does using an MSO have for protocol analysis?

Mixed-signal oscilloscopes (MSOs) make great choices for protocol analysis for several reasons. First, they free up analogue scope channels for viewing other system activity (see Fig. 3). Second, when viewing more than one serial bus, MSOs offer additional channels unlike digital signal oscilloscopes that max out at four channels. Third, some vendors have more standard MSO acquisition memory than is available on scope channels, enabling capture of additional packets when the MSO's digital channels are used rather than analogue channels.

Many vendors do not support segmented memory with MSO digital channels. This means that MSO channels cannot provide protocol decode when segmented memory mode is utilised. Check whether your scope vendor supports segmented memory on the MSO channels.

Adding protocol analysis capabilities to a scope enables teams to debug a wider range of issues faster. Evaluating both specific protocol apps, and the scope's underlying ability to effectively perform packet-based triggering and decode will result in selection of the scope that best fits your team's need. ●

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